In [ ]:
```python
#Loops
```

In [1]:
```python
i = 1
while i<=5:
    print('Data Science')
    i = i+1
```

```
Data Science
Data Science
Data Science
Data Science
Data Science
```

In [2]:
```python
i = 5
while i>=1:
    print('Data Science')
    i = i-1
```

```
Data Science
Data Science
Data Science
Data Science
Data Science
```

In [3]:
```python
i = 1
while i<=5:
    print('Data Science :',i)
    i = i+1
```

```
Data Science : 1
Data Science : 2
Data Science : 3
Data Science : 4
Data Science : 5
```

In [4]:
```python
i = 5
while i>=1:
    print('Data Science :',i)
    i = i-1
```

```
Data Science : 5
Data Science : 4
Data Science : 3
Data Science : 2
Data Science : 1
```

In [5]:

```python
i = 1
while i<=5:
    print('Data Science')
    j = 1
    while j<=4:
        print('technology')
        j = j+1
    i = i+1
    print()
```

```
Data Science
technology
technology
technology
technology

Data Science
technology
technology
technology
technology

Data Science
technology
technology
technology
technology

Data Science
technology
technology
technology
technology

Data Science
technology
technology
technology
technology
```

In [6]:
```
i = 1
while i<=5:
    print('Data Science', end = " ")
    j = 1
    while j<=4:
        print('technology', end=" ")
        j = j+1
    i = i+1
    print()
```

```
Data Science technology technology technology technology
Data Science technology technology technology technology
Data Science technology technology technology technology
Data Science technology technology technology technology
Data Science technology technology technology technology
```

In [7]:

```python
i = 1
while i<=2:
    j = 0
    while j<=2:
        print(i*j, end=" ")
        j = j+1
    print()
    i = i+1
```

```
0 1 2
0 2 4
```

In [8]:

```python
i = 1
while i<=4:
    j = 0
    while j<=3:
        print(i*j, end=" ")
        j = j+1
    print()
    i += 1
```

```
0 1 2 3
0 2 4 6
0 3 6 9
0 4 8 12
```

In [9]:

```python
i = 10
while i<=4:
    j = 0
    while j<=3:
        print(i*j, end=" ")
        j = j+1
    print()
    i += 1
```

In [10]:
```
i = 10
while i<=4:
    j = 0
    while j<=3:
        print(i*j, end=" ")
        j = j+1
    print()
    i += 1
else:
    print('Condition is not matched.')
```

```
Condition is not matched.
```

In [11]:
```
name = 'nit'
for i in name:
    print(i)
```

```
n
i
t
```

In [12]:
```
name1 = [1,3.5,'hallo']
for i in name1:
    print(i)
```

```
1
3.5
hallo
```

In [13]:
```
for i in range(5):
    print(i)
```

```
0
1
2
3
4
```

In [14]:
```
for i in range(10,50,5):
    print(i)
```

```
10
15
20
25
30
35
40
45
```

In [16]:
```python
#Print value divisible by 5 (Table 5)
for i in range(1,51):
    if i%5 == 0:
        print(i)
```

```
5
10
15
20
25
30
35
40
45
50
```

In [17]:
```python
#Print values not divisible by 5
for i in range(1,51):
    if i%5 != 0:
        print(i)
```

In [18]:
```
#Print value divisible by 3 (Table 3)
for i in range(1,31):
    if i%3 == 0:
        print(i)
```

```
3
6
9
12
15
18
21
24
27
30
```

In [19]:
```
for i in range(1,11):
    print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

In [20]:
```
#Print 5 numbers from range 1 to 10
for i in range(1,11):
    if i ==6:
        break
    print(i)
```

```
1
2
3
4
5
```

In [22]:
```
#Skip 4 and print rest of the loop
for i in range(1,11):
    if i ==4:
        continue
    print(i)
```

```
1
2
3
5
6
7
8
9
10
```

In [23]:
```
for i in range(1,11):
```

```
  Cell In[23], line 1
    for i in range(1,11):
                         ^
_IncompleteInputError: incomplete input
```

In [ ]:
```
#The above is error. Hence, pass is used.
```

In [24]:
```
for i in range(1,11):
    pass
```

In [27]:
```
#Vending machine
x = int(input("How many chocolates you want ?"))
i = 1
while i<=x:
    print("Chocolate: ", i)
    i = i+1
```

```
Chocolate:  1
Chocolate:  2
Chocolate:  3
```

In [29]:
```
#Say available chocolates with vending machine is 5
available_chocolate = 5
x = int(input("How many chocolates you want ?"))
```

```python
i = 1
while i<=x:
    if i>available_chocolate:
        print("Out of Stock")
        break
    print("Chocolate: ", i)
    i = i+1
print("Bye for now")
```

```
Chocolate:  1
Chocolate:  2
Chocolate:  3
Chocolate:  4
Chocolate:  5
Out of Stock
Bye for now
```

In [30]:

```python
#Print 1 tp 50. Don't print divisible by 3 or 5 numbers
for i in range(1,51):
    if i%3 == 0 or i%5 == 0:
        continue
    print(i)
```

```
1
2
4
7
8
11
13
14
16
17
19
22
23
26
28
29
31
32
34
37
38
41
43
44
46
47
49
```

In [31]:

```
nums = [12,15,18,21,26,30,40]
for num in nums:
    if num%5 == 0:
        print(num)
```

```
15
30
40
```

In [32]:

```
nums = [12,15,18,21,26,30,40]
for num in nums:
    if num%5 == 0:
        print(num)
        break
```

```
15
```

In [34]:
```python
nums = [12,21,26]
for num in nums:
    if num%5 == 0:
        print(num)
```

In [ ]:
```python
#The above has no output. Hence, for else is used
```

In [37]:
```python
nums = [12,15,21,26]
for num in nums:
    if num%5 == 0:
        print(num)
else:
    print("Number not found")
```

```
15
Number not found
```

In [ ]:
```python
#The above is not correct. Hence, below
```

In [36]:
```python
nums = [12,18,21,26,15]
for num in nums:
    if num%5 == 0:
        print(num)
        break
else:
    print("Number not found")
```

```
15
```

In [ ]:
```python

```